# Bayesian tree ensembles for heterogeneous treatment effects on survival from randomised and real-world data

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**Abstract**

We develop a Bayesian nonparametric framework to estimate heterogeneous treatment effects on survival outcomes by combining a randomised controlled trial and real-world data. We relax the unconfoundedness assumption on the real-world data. We model the survival time with an accelerated failure time decomposition into a baseline prognosis, a treatment effect, and a confounding function. The confounding function absorbs the confounding bias in the real-world data. Each component receives a Bayesian tree ensemble prior. We share the baseline prognosis across the two sources while allowing for between-source heterogeneity. We model the error distribution nonparametrically with a hierarchical Dirichlet process mixture. We allow for both right- and interval-censored data. A simulation study shows efficiency gains over a trial-only analysis across varying levels of confounding and between-source heterogeneity. The Bayesian fusion forest also improves on a flexible black-box competitor in both accuracy and calibration. We combine the ACTG 175 trial with the Multicenter AIDS Cohort Study to study the effect of combination antiretroviral therapy for HIV. The fusion identifies a benefit for nearly every patient where the trial alone is inconclusive.

**Keywords:** Heterogeneous treatment effects; data fusion; real-world evidence; unmeasured confounding; Bayesian additive regression trees; survival analysis.

**Alternative titles:**

- Bayesian nonparametric fusion of randomised and real-world survival data for heterogeneous treatment effects
- Heterogeneous treatment effects on survival from randomised and real-world data under unmeasured confounding
- A confounding-function approach to combining trials and real-world data for heterogeneous survival effects
- Bayesian data fusion for heterogeneous survival treatment effects

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# 1 Introduction

Precision medicine uses heterogeneous treatment effects (HTE) to guide clinical decisions. Researchers estimate the HTE from either a randomised controlled trial (RCT) or observational real-world data (RWD). Randomised trials are the gold standard for causal effect estimation, but they are underpowered for subgroup effects and often have limited follow-up. Real-world data from registries and cohorts are larger and often follow patients for longer. Regulators increasingly accept them as evidence. Unmeasured confounding can bias treatment effect estimates from observational data. We combine the two data sources. We identify the effect from the trial and borrow power and longer follow-up from the real-world data. Our motivating example is HIV antiretroviral therapy. The treatment effect varies with baseline CD4 count, viral load, and age. We fuse the ACTG 175 trial (Hammer et al., 1996) with the Multicenter AIDS Cohort Study (Kaslow et al., 1987). The cohort followed participants over more than 25 years with periodic visits.

The fusion of a trial with real-world data raises three challenges for survival outcomes. The first is unmeasured confounding in the real-world data. Treatment is not randomised, and assignment depends on factors the measured covariates do not fully capture. The no-unmeasured-confounders assumption is untestable and often violated in practice (VanderWeele and Ding, 2017). The second is heterogeneity between the two data sources. Even when the treatment effect transports across sources, the baseline prognosis often does not. The trial and observational populations typically differ in eligibility, follow-up, and case mix. A naive analysis confounds this baseline difference with the treatment effect. This heterogeneity is not confined to the mean: differences in measurement protocol and follow-up quality also reshape the error distribution across sources. The third is the survival outcome. Right censoring arises when the event has not occurred by the end of follow-up, so the survival time is known only to exceed the censoring time. The real-world data may also be interval-censored, with events recorded only at periodic visits.

Unmeasured confounding can be addressed through a confounding function. The confounding function measures the confounding bias in the real-world data. It was introduced as a sensitivity-analysis device within a single study (Robins et al., 2000). The confounding function can be estimated by combining a randomised and an observational study. Kallus et al. (2018) learn the confounding function from the RCT and debias the observational treatment effect. Yang et al. (2025) develop a semiparametric-efficient estimator that jointly estimates the treatment effect and the confounding function, and derive a test for unmeasured confounding. The elastic integrative estimator of Yang et al. (2023) uses that test to decide whether to borrow from the real-world data, and discards it when confounding is detected. Two frequentist methods extend the approach to survival outcomes. Ye et al. (2025) work under an accelerated failure time model with right censoring and high-dimensional covariates, and treat the confounding function as a sparse component selected jointly with the treatment effect. Mao et al. (2025) target the conditional restricted mean survival time under right censoring.

Bayesian methods combine data sources by borrowing information through the prior. The power prior of Chen and Ibrahim (2000) raises the external likelihood to a fractional power that controls the borrowing. The meta-analytic-predictive prior of Neuenschwander et al. (2010) makes the borrowing hierarchical and adaptive to between-source heterogeneity. Later variants tie the borrowing to the between-source discrepancy through commensurate priors (Hobbs et al., 2012). These methods all target a marginal or control-arm effect. More recent literature targets heterogeneous effects directly. Zhou and Ji (2021) pool trial and external outcome surfaces with Bayesian additive regression trees, but do not allow for unmeasured confounding in the external source. Dimitriou et al. (2026) develop a multi-task Gaussian process with a data-adaptive borrowing parameter. None of these Bayesian methods targets heterogeneous treatment effects on survival outcomes under unmeasured confounding.

Bayesian additive regression trees (Chipman et al., 2010) are a popular method for causal machine learning (Hill, 2011). They are supported by theoretical guarantees (Ročková and van der Pas, 2020) and have strong empirical performance in both estimation accuracy and uncertainty quantification (Dorie et al., 2019; Thal and Finucane, 2023; Kabata et al., 2026). The uncertainty quantification is intrinsic to the model and follows directly from the posterior distribution. Black-box predictors such as (deep) neural networks are flexible alternatives, but do not provide intrinsic uncertainty quantification and do not separate the causal effect from confounding bias. We will show that a structured approach tailored to the problem at hand and its statistical constraints outperforms a generic black-box competitor in both accuracy and calibration.

We develop a Bayesian nonparametric framework to estimate heterogeneous treatment effects on survival outcomes. We target the acceleration factor, a causally interpretable estimand on the survival-time scale. We fuse a randomised trial with real-world data without assuming the observational source is unconfounded. A confounding function absorbs the bias from unmeasured confounding in the real-world data. We accommodate right- and interval-censored data, and exploit the longer follow-up of the real-world source for robustness. A hierarchical Dirichlet process mixture models the error distribution nonparametrically and lets it vary across sources.

# 2 Methodology

## 2.1 Causal framework

We consider data from two sources: a randomised controlled trial (RCT) and a real-world data study (RWD). The data-source indicator $S \in \{0, 1\}$ takes the value $S = 1$ for RCT observations and $S = 0$ for RWD observations. The RCT contributes $n_1$ independent observations and the RWD contributes $n_0$ independent observations. We observe a vector of pre-treatment covariates $X \in \mathbb{R}^p$ and a binary treatment $A \in \{0, 1\}$ for each individual. We denote the random variables without the subject index $i$ in general. The outcome of interest is the nonnegative survival time $T$, subject to censoring. We allow two types of censoring: right censoring, and interval censoring. For each subject, we observe a pair $(L, R)$ with $0 \leq L \leq R \leq \infty$ satisfying $T \in [L, R]$, together with a censoring-type indicator $\delta \in \{0, 1, 2\}$. We use the convention that $R = \infty$ stands for $T \in [L, \infty)$. These correspond to exact observation ($\delta = 1$: $L = R = T$), right censoring ($\delta = 0$: $L = C$ and $R = \infty$, for some censoring time $C$), and interval censoring ($\delta = 2$: $0 \leq L < R < \infty$). The standard right-censored setup is recovered when $\delta \in \{0, 1\}$ for every subject. We write $\mathcal{C}$ for the inspection and censoring process underlying $(L, R, \delta)$, and refer to Sun (2006) for details. We adopt the potential-outcomes framework (Rubin, 1974). For $a \in \{0, 1\}$, the potential survival time $T(a)$, the censoring process $\mathcal{C}(a)$, and the observation $(L(a), R(a), \delta(a))$ are the quantities that would be observed if treatment were set to $a$.

We estimate the conditional average treatment effect (CATE):

$$\tau(x) = \mathbb{E}[\log T(1) - \log T(0) \mid X = x]. \tag{1}$$

The CATE describes how the treatment effect varies across covariates. The corresponding estimand on the multiplicative time scale is the *acceleration factor* $\exp(\tau(x))$. The acceleration factor admits a direct interpretation on the survival-time scale: at every quantile of the survival distribution, treatment rescales the survival time by a constant multiplicative factor. This time-scaling is more intuitive in a clinical setting than the relative change in event rate conveyed by a hazard ratio (Swindell, 2009). The acceleration factor also enjoys causal properties that the hazard ratio lacks. Brathovde et al. (2026) show that the observed acceleration factor identifies its causal counterpart under exchangeability and consistency. This identification continues to hold under unmeasured frailty and treatment-effect heterogeneity. The hazard ratio loses its causal interpretation in those same settings, because conditioning on survival induces a built-in selection bias (Hernán, 2010). Moreover, the acceleration factor is collapsible: the marginal and conditional acceleration factors coincide when the treatment effect is homogeneous, regardless of the distribution of unmeasured baseline risk (Crowther et al., 2023).

We impose the following assumptions on the causal structure and the censoring mechanism.

**Assumption 1** (Consistency). $T = A\,T(1) + (1-A)\,T(0)$.

**Assumption 2** (Unconfoundedness of the RCT). $T(a) \perp\!\!\!\perp A \mid X, S = 1$ *for each* $a \in \{0, 1\}$.

**Assumption 3** (Positivity). $0 < \mathbb{P}(A = 1 \mid X = x, S = s) < 1$ *for each* $s \in \{0, 1\}$ *and all* $x$ *in the support of* $X \mid S = s$.

**Assumption 4** (Cross-source transportability). $\mathbb{E}[\log T(1) - \log T(0) \mid X, S = 1] = \mathbb{E}[\log T(1) - \log T(0) \mid X, S = 0]$.

**Assumption 5** (Conditionally non-informative censoring). $T(a) \perp\!\!\!\perp \mathcal{C}(a) \mid X, A, S$ *for each* $a \in \{0, 1\}$.

Our identification strategy relaxes unconfoundedness in the RWD and retains transportability of the CATE across sources. Assumption 2 holds by design in a randomised trial. Treatment in the RWD may depend on confounders not contained in $X$. The resulting bias is absorbed by the confounding function $c(x)$ introduced in the next section. We assume the CATE to be time-invariant by not explicitly letting it depend on time. Other approaches that combine evidence from multiple sources take the opposite route: they allow source-specific CATEs but require unconfoundedness within each source (Stuart et al., 2011; Dahabreh et al., 2019; Shyr et al., 2025). The two relaxations cannot coexist in our setting. Without unconfoundedness of the RWD, the CATE and $c(x)$ are not separately identifiable unless the CATE transports across sources. We therefore allow for confounding in the real-world data at the explicit cost of retaining Assumption 4.

### 2.2 Accelerated failure time decomposition

We model the conditional log survival time jointly across the trial and the real-world data. In the real-world data, the treated-versus-control contrast generally differs from the causal treatment effect. Treatment assignment depends on confounders outside $X$, which bias the observed contrast. We define the *confounding function* $c(x)$ as:

$$c(x) \;=\; \mathbb{E}[\log T \mid X = x,\, A = 1,\, S = 0] \;-\; \mathbb{E}[\log T \mid X = x,\, A = 0,\, S = 0] \;-\; \tau(x). \tag{2}$$

The confounding function captures the difference between the RWD treated-versus-control contrast and the true causal effect. The source indicator $S$ ties the two sources into one model. A baseline prognosis and the causal effect enter in both sources, while the confounding bias enters only in the real-world data.

**Proposition 1** (Accelerated failure time decomposition). *Under Assumptions 1–4, the conditional log survival time satisfies:*

$$\mathbb{E}[\log T \mid A,\, X,\, S] \;=\; m_0(X, S) \;+\; \tau(X)\,A \;+\; (1 - S)\,A\,c(X), \tag{3}$$

*where* $m_0(X, S) := \mathbb{E}[\log T \mid A = 0,\, X,\, S]$ *is the baseline log survival time.*

The baseline prognosis $m_0(X, S)$ is the survival a patient would have without treatment, which can differ between the two sources. The confounding term carries the factor $(1 - S)$, so it acts only in the real-world data.

**Proposition 2** (Identification)**.** *Under Assumptions 1–4:*

(i) *The CATE is identified from the RCT alone:*

$$\tau(x) \;=\; \mathbb{E}[\log T \mid X = x,\, A = 1,\, S = 1] \;-\; \mathbb{E}[\log T \mid X = x,\, A = 0,\, S = 1]. \tag{4}$$

(ii) *Given $\tau$, the confounding function is identified from the RWD.*

(iii) *Neither $\tau$ nor $c$ is identified from the RWD alone; the RWD identifies only the composite $\tau(x) + c(x)$.*

Proofs of Propositions 1 and 2 are in Supplementary Materials S.1. The same identification results hold for the acceleration factor $\exp(\tau(x))$ by continuity of the exponential.

Propositions 1 and 2 concern the population conditional means $\mathbb{E}[\log T \mid A, X, S]$ of the log transformed survival time. Assumptions 1–4 identify the causal quantities from these conditional means. Assumption 5 on the censoring mechanism plays a separate role: it lets us recover the conditional means from the censored observations $(L, R, \delta)$. The censoring assumption enters the likelihood and the data augmentation in the sampler, not the causal identification argument.

We model the log survival time with the accelerated failure time (AFT) specification implied by the decomposition:

$$\log T \;=\; m_0(X, S) \;+\; A\,\tau(X) \;+\; (1 - S)\,A\,c(X) \;+\; \varepsilon. \tag{5}$$

We assume the errors are independent across subjects and have mean zero. We impose no parametric form on their distribution and allow it to differ across sources through a hierarchical prior. The regression functions $m_0$, $\tau$ and $c$ retain the causal meanings established above. We assign each function a separate Bayesian tree ensemble prior.

## 2.3 Nonparametric error distribution

We model the error $\varepsilon$ in (5) with a hierarchical Dirichlet process mixture of normals (HDPM). The prior is source-specific: $\varepsilon \mid S = s \sim F_s$, where $F_s$ is the distribution of the error term in source $s$. We let $F_0$ and $F_1$ differ flexibly while sharing a common set of mixture components across sources. Existing data-fusion methods typically assume $F_0 = F_1$, for example via a single Gaussian likelihood. This assumption is unsatisfactory in our setting. Randomised trials and real-world data sources typically differ in measurement protocol, follow-up quality, and the prevalence of extreme outcomes. These differences plausibly enter the error distribution rather than the structural mean. But the two error distributions are unlikely to be entirely unrelated. A model that treats them as fully independent forfeits useful pooling, particularly when one source is small.

Conditional on the source indicator $S = s$, we model the error as a location mixture of Gaussians with source-specific scale $\sigma_s > 0$ and mixing measure $G_s$:

$$\varepsilon \mid S = s,\, G_s,\, \sigma_s \;\sim\; \int \frac{1}{\sigma_s}\,\phi\!\left(\frac{w - \theta}{\sigma_s}\right) dG_s(\theta). \tag{6}$$

Here $\phi$ is the standard normal density and $\theta$ is the location of a mixture component. The right-hand side gives the density of $F_s$, evaluated at the argument $w$. Gaussian-mixture priors of this form have been used for AFT residuals in single-source BART models (Henderson et al., 2020). The mixing measures share a latent structure through the hierarchical Dirichlet process (Teh et al., 2006) with a top-level random measure $G^*$:

$$G^* \sim \mathrm{DP}(\gamma, H), \qquad G_s^* \mid G^*, M_s \sim \mathrm{DP}(M_s, G^*), \qquad s \in \{0, 1\}, \tag{7}$$

with base distribution $H = \mathcal{N}(0, \sigma_\theta^2)$, top-level concentration parameter $\gamma$, and source-specific concentration parameters $M_s$. The top-level measure is discrete and admits a stick-breaking representation:

$$G^* = \sum_{k=1}^{\infty} \beta_k \, \delta_{\theta_k^*}, \qquad \theta_k^* \overset{\text{iid}}{\sim} H, \qquad \beta_k = u_k \prod_{l<k} (1 - u_l), \qquad u_k \overset{\text{iid}}{\sim} \mathrm{Beta}(1, \gamma). \tag{8}$$

Each source-specific measure $G_s^*$ is supported on the same atoms $\{\theta_k^*\}_{k \geq 1}$ as $G^*$, with its own weight vector $\boldsymbol{\pi}_s = (\pi_{sk})_{k \geq 1}$. The shape of each error distribution $F_s$ is therefore parameterised by a shared set of mixture components and a source-specific weighting over them.

To retain the interpretation $m_0$, $\tau$, and $c$ as the conditional-mean components defined in (3), each $F_s$ must have mean zero. We follow Yang et al. (2010) and obtain centred mixing measures by subtracting the source-specific mean of the unconstrained measure:

$$\mu_s = \int \theta \, dG_s^*(\theta) = \sum_{k=1}^{\infty} \pi_{sk} \, \theta_k^*, \qquad G_s = \sum_{k=1}^{\infty} \pi_{sk} \, \delta_{\theta_k^* - \mu_s}. \tag{9}$$

By construction $\mathbb{E}[\varepsilon \mid S = s, G_s, \sigma_s] = 0$. The unconstrained atoms $\theta_k^*$ remain shared across sources, while the centred atoms $\theta_k^* - \mu_s$ are source-specific through the shifts $\mu_0$ and $\mu_1$. We place a scaled-inverse-$\chi^2$ prior $\sigma_s^2 \sim \nu\lambda/\chi_\nu^2$ on the per-source residual scales, with degrees of freedom $\nu$ and scale $\lambda$. We set $\nu = 3$ following the BART default and calibrate $\lambda$ to the empirical residual variance (Chipman et al., 2010).

## 2.4 Overview of BART

Bayesian additive regression trees (Chipman et al., 2010) provide a flexible nonparametric prior over an unknown function $f : \mathbb{R}^p \to \mathbb{R}$. The prior represents $f$ as a sum of $J$ regression trees:

$$f(x) = \sum_{j=1}^{J} g(x; \mathcal{T}_j, \mathcal{H}_j), \tag{10}$$

where each $\mathcal{T}_j$ is a binary tree with $b_j$ terminal nodes. Each interior splitting rule has the form $\{x_\ell \leq x^*\}$ for some covariate index $\ell$ and splitting value $x^*$. We write $\mathcal{H}_j = \{h_{j,1}, \ldots, h_{j,b_j}\}$ for the step heights at those terminal nodes. The function $g(x; \mathcal{T}_j, \mathcal{H}_j)$ routes $x$ through the splits of $\mathcal{T}_j$ to a terminal node $r$ and returns the step height $h_{j,r}$ at that node.

The prior on $(\mathcal{T}_1, \mathcal{H}_1), \ldots, (\mathcal{T}_J, \mathcal{H}_J)$ is independent across trees. It decomposes into a prior on the tree-structure $\mathcal{T}_j$ and a prior on the step heights $\mathcal{H}_j$ given the tree structure $\mathcal{T}_j$. The structure of each $\mathcal{T}_j$ follows the recursive splitting process of Chipman et al. (1998). A node at depth $d$ is non-terminal with probability $\alpha(1 + d)^{-\beta}$. The hyperparameters $\alpha$ and $\beta$ regularise tree depth. Conditional on the tree, the splitting variable $x_\ell$ at each interior node is drawn uniformly from the available covariates. The splitting value $x^*$ is drawn uniformly from its observed values. The step heights receive independent conjugate normal priors $h_{j,r} \mid \mathcal{T}_j \sim \mathcal{N}(0, \sigma_h^2)$ with

$\sigma_h = k/(2\sqrt{J})$ and $k > 0$ the step-height scale. A horseshoe prior on the step heights is a more suitable alternative in high-dimensional settings (Jacobs et al., 2025). Chipman et al. (2010) centre and rescale the response, then choose $k$ so that the implied prior on $f$ assigns circa 95% of its mass to the observed response range. We denote the BART prior with these hyperparameters by $\mathrm{BART}(J, k, \alpha, \beta)$. The four arguments are the number of trees $J$, the step-height scale $k$, and the tree-structure hyperparameters $\alpha$ and $\beta$.

### 2.5 Prior specification

We place a separate Bayesian additive regression tree prior on each component of the decomposition (3): the baseline prognosis $m_0$, the treatment effect $\tau$, and the confounding function $c$. Each component carries an independent causal meaning (Proposition 1), so each prior acts directly on an interpretable quantity. An undifferentiated BART on the full conditional mean $\mathbb{E}[\log T \mid A, X, S]$ would reach the priors on $\tau$ and $c$ only indirectly. The induced prior on a contrast then depends on incidental features such as the dimension and distribution of $X$ (Hahn et al., 2020). It also cannot encode the between-source borrowing and the confounding shrinkage that the problem calls for. The decomposition sacrifices nothing in expressiveness, since any conditional mean of the form (3) stays in the support. It only changes which functions the prior deems likely. This separate-forest construction generalises the prognostic-treatment split of the Bayesian causal forest (Hahn et al., 2020) to two data sources and an explicit confounding function. We tune each prior through its tree-structure hyperparameters and its ensemble size. We order the regularisation along a ladder, from the flexible baseline prognosis to the strongly regularised confounding function.

The baseline prognosis $m_0(X, S)$ is the first component, and it may differ between the two sources. We split it into a shared component and a source-specific deviation:

$$m_0(X, S) \;=\; m_0^{\mathrm{sh}}(X) + (1 - S)\, d(X), \tag{11}$$

where $m_0^{\mathrm{sh}}$ is shared across both sources and the deviation $d$ is active only in the real-world data. The deviation $d$ is the difference between the two source-specific baselines. We place a separate BART prior on $m_0^{\mathrm{sh}}$ and on $d$, and we give $d$ a mean-zero prior. The prior then centres on equality of the two source baselines, which encodes a preference for borrowing. A single BART on $m_0(X, S)$ with $S$ as a binary covariate would not encode it. We read this split as a meta-analytic-predictive prior on the prognostic function (Neuenschwander et al., 2010). The shared forest is the common borrowing target, and the mean-zero deviation forest measures how far the real-world baseline drifts from it. This reading is the argument for the separate deviation forest: two independent baselines would borrow nothing between the sources, while the mean-zero deviation pulls the real-world baseline towards the trial baseline by an amount set by its step-height scale $k_d$. A small $k_d$ shrinks the deviation towards zero for strong borrowing, and a large $k_d$ lets the real-world baseline depart (Supplementary Materials S.2).

We let $m_0^{\mathrm{sh}}$ be the most flexible forest in the model. We use the tree-structure hyperparameters $(\alpha_{\mathrm{sh}}, \beta_{\mathrm{sh}}) = (0.95, 2)$ and $J_{\mathrm{sh}} = 200$ trees, so the shared baseline can accommodate complex disease-intrinsic structure. We use the same $(\alpha_d, \beta_d) = (0.95, 2)$ for $d$ but only $J_d = 50$ trees. The smaller ensemble suffices because $d$ only absorbs residual differences in protocol and patient population. The deviation keeps the same permissive tree-structure prior, so it can track covariate-dependent differences between the sources.

The treatment effect $\tau$ is the second component, and we regularise it more strongly than the baseline. We place a depth-penalised BART prior on $\tau$ to discourage higher-order interactions in $X$. We set $\beta_\tau = 3$ and keep $\alpha_\tau = 0.95$, so the prior mass concentrates on trees of depth one or two. We reduce the ensemble to $J_\tau = 100$ trees, which further strengthens the regularisation. At the no-split extreme the treatment forest reduces to a constant, a homogeneous treatment effect. The prior thus shrinks the conditional average treatment effect towards homogeneity. The stronger regularisation reflects the expectation that the treatment-effect surface is simpler than the baseline prognosis.

The confounding function $c$ is the third component, and we regularise it most strongly of all. We regularise it through both the root-split probability and the depth penalty. We set the base splitting probability $\alpha_c = 0.25$ and the depth parameter $\beta_c = 3$, and use $J_c = 50$ trees. The low base probability discourages splits at the root. The depth penalty discourages deep interactions when splits do occur. The settings $(\alpha_c, \beta_c) = (0.25, 3)$ express a strong prior preference for confounding that varies little with the covariates, close to a uniform bias.

We standardise the log survival time before fitting by the centring and scaling constants of a preliminary log-normal AFT fit. This rescales the response to a unit-variance scale. Each forest carries its own prior step-height scale $k_f$, indexed by $f \in \{\mathrm{sh}, d, \tau, c\}$ for the shared baseline $m_0^{\mathrm{sh}}$, the deviation $d$, the treatment effect $\tau$, and the confounding function $c$. The step-height scales calibrate the prior magnitude of each ensemble to the scale of the standardised response. We set $k_f = 1$ for every forest by default. We halve it for the treatment effect, $k_\tau = 1/2$, which shrinks the magnitude of $\tau$ on top of the structural regularisation already placed on it. We tune further by cross-validation.

We summarise the full hierarchical model:

$$
\begin{aligned}
\log T_i &= m_0^{\mathrm{sh}}(X_i) + (1 - S_i)\, d(X_i) + A_i\, \tau(X_i) + (1 - S_i)\, A_i\, c(X_i) + \varepsilon_i, \\
m_0^{\mathrm{sh}} &\sim \mathrm{BART}(200,\, k_{\mathrm{sh}},\, 0.95,\, 2), \\
d &\sim \mathrm{BART}(50,\, k_d,\, 0.95,\, 2), \\
\tau &\sim \mathrm{BART}(100,\, k_\tau,\, 0.95,\, 3), \\
c &\sim \mathrm{BART}(50,\, k_c,\, 0.25,\, 3), \\
\varepsilon_i \mid S_i = s,\, G_s,\, \sigma_s &\sim \int \sigma_s^{-1}\, \phi\!\left(\tfrac{w-\theta}{\sigma_s}\right) dG_s(\theta), \\
G_s &= \textstyle\sum_k \pi_{sk}\, \delta_{\theta_k^* - \mu_s}, \quad s \in \{0, 1\}, \\
G_s^* \mid G^*,\, M_s &\sim \mathrm{DP}(M_s,\, G^*), \quad s \in \{0, 1\}, \\
G^* \mid \gamma &\sim \mathrm{DP}(\gamma,\, H), \qquad H = \mathcal{N}(0,\, \sigma_\theta^2), \\
\gamma &\sim \mathrm{Gamma}(a_\gamma,\, b_\gamma), \\
M_s &\sim \mathrm{Gamma}(a_M,\, b_M), \quad s \in \{0, 1\}, \\
\sigma_s^2 &\sim \nu\lambda/\chi^2_\nu, \quad s \in \{0, 1\}.
\end{aligned}
$$

We calibrate the base-measure variance $\sigma_\theta^2$ rather than assign it a prior. Supplementary Materials S.3 give the hyperparameter values.

We refer to this model as the *Bayesian fusion forest*.

## 2.6 Posterior inference

We sample from the posterior via a blocked Gibbs sampler. We update the BART forests $m_0^{\mathrm{sh}}$, $d$, $\tau$, and $c$ in turn against the partial residual. The partial residual subtracts the other forests and the current draw of the residual mixture from the augmented outcome. Each update reduces to a standard BART update on a Gaussian working response (Chipman et al., 2010). We augment the censored observations (Tanner and Wong, 1987). The HDPM error block is updated by a separate Gibbs step over its cluster assignments, atoms, mixture weights, and concentration parameters. We give the full sampler in the Supplementary Materials.

We compute average treatment effects by marginalising the conditional effect $\tau(\cdot)$ against a target covariate distribution. We take the target to be the union of the two source studies, with covariate distribution $F_X = \pi_0 F_X^0 + \pi_1 F_X^1$. Here $F_X^s$ is the covariate distribution within source $s \in \{0, 1\}$ and $\pi = (\pi_0, \pi_1)$ lies on the unit simplex. The posterior of the marginal effect carries uncertainty from three sources: the posterior over $\tau$, the within-source covariate distributions $F_X^s$, and the mixing fractions $\pi$. We propagate all three jointly by a hierarchical Bayesian bootstrap. The choice of $\pi$ encodes the analyst's target population. Natural deterministic choices include the empirical fractions $\pi_s = n_s/n$ and equal weighting $\pi = (1/2, 1/2)$. The source-specific limits $\pi = (1, 0)$ and $\pi = (0, 1)$ target the population of the RWD and RCT respectively. We may instead place a Dirichlet prior $\pi \sim \text{Dirichlet}(\alpha)$ on the mixing fractions, with concentration parameter $\alpha = (\alpha_0, \alpha_1)$. This propagates uncertainty about the target composition. The choice $\alpha = (n_0, n_1)$ centres the prior on the empirical fractions and approximately recovers a pooled Bayesian bootstrap. We recommend this as the default. The choice $\alpha = (1, 1)$ gives a flat prior on the simplex. We draw mixing fractions $\pi^{(b)} \sim \text{Dirichlet}(\alpha)$ for each posterior draw $\tau^{(b)}(\cdot)$ of the conditional effect. We draw within-source weights $w^{(s,b)} \sim \text{Dirichlet}(\mathbf{1}_{n_s})$ over the observations of source $s$ independently for $s \in \{0, 1\}$. The corresponding draw of the average treatment effect is:

$$\bar{\tau}^{(b)} = \sum_{s\in\{0,1\}} \pi_s^{(b)} \sum_{i:S_i=s} w_i^{(s,b)} \, \tau^{(b)}(X_i). \tag{12}$$

The within-source weights are the Bayesian bootstrap (Rubin, 1981) applied to each $F_X^s$.

We can summarise each fitted forest by projecting it on a lower-dimensional model. The posterior draws of $m_0^{\text{sh}}$, $d$, $\tau$, and $c$ retain the causal meaning established in Proposition 1. We can therefore summarise any one of them in isolation. We project a fitted forest onto a simple second-stage model since a full posterior over a nonparametric surface is hard to communicate. This follows the posterior-summarisation approach of Woody et al. (2021) and the two-stage logic of the Virtual Twins method (Foster et al., 2011). The first stage is the fitted forest. The second is a low-complexity model fit to the posterior-mean surface of that forest, such as a single regression tree or a linear projection. We propagate posterior uncertainty by fixing the second-stage structure at its posterior-mean fit and recomputing its parameters on every posterior draw.

# 3 Simulation study

We compare the Bayesian fusion forest on the combined data sources with a Bayesian causal forest on the RCT and RWD alone. We assess its robustness to unmeasured confounding and to between-source heterogeneity, benchmark it against a flexible deep-learning competitor, and study its behaviour as the covariate dimension grows.

## 3.1 Simulation setup

The general setup is common to all three experiments below. The trial contributes $n_1 = 150$ observations and the real-world data $n_0 = 350$. We draw $p = 10$ covariates $X \sim \mathcal{N}(0, \Sigma)$ in both sources, with an AR(1) correlation $\Sigma_{ij} = \rho^{|i-j|}$, $\rho = 0.3$. Five covariates are active and five are noise. We generate the outcome:

$$\begin{aligned} \log T_{\mathrm{RCT}} &= m_0(x) + A\,\tau(x) + \varepsilon_{\mathrm{RCT}}, \\ \log T_{\mathrm{RWD}} &= m_0(x) + \lambda_d\, d(x) + A\,\tau(x) + \lambda_u\, A\, U + \varepsilon_{\mathrm{RWD}}, \end{aligned} \tag{13}$$

with components:

$$m_0(x) = 4x_1 - 2x_2x_3 + x_4^2, \quad \tau(x) = 1/2 + 2x_1 - x_2^2, \quad d(x) = 2x_4 - x_5. \tag{14}$$

The function $\tau$ is the true CATE on the log-time scale. We randomise treatment in the trial, $A \sim \mathrm{Bernoulli}(1/2)$, and let real-world treatment depend on an unmeasured confounder $U \sim \mathrm{Uniform}(0,1)$ through $A \sim \mathrm{Bernoulli}(\mathrm{expit}(x_1 + U))$. The confounder enters the outcome only in the treated arm. The factor $\lambda_d$ controls the between-source heterogeneity in baseline prognosis and $\lambda_u$ the strength of the unmeasured confounding. We model the trial error as $\varepsilon_{\mathrm{RCT}} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 3/4$, and draw the real-world error from a standardised Gumbel law: $\varepsilon_{\mathrm{RWD}} = \frac{\sqrt{6}}{\pi}\gamma_{\mathrm{E}} - G$ with $G \sim \mathrm{Gumbel}(0, \sqrt{6}/\pi)$, which has mean zero, unit variance, and is skewed. Trial and real-world event times are conditionally log-normal and Weibull, respectively. We right-censor the trial by an exponential censoring time tuned to circa 35% censoring. We interval-censor the real-world data at eight inspection times. Events after the last inspection are right-censored, circa 20% of the real-world observations.

We fit the Bayesian fusion forest to the combined data with default parameters. The single-source baseline is an accelerated failure time Bayesian causal forest (Jacobs, 2026), fitted to the trial and to the real-world data separately. We run 1000 Monte Carlo replications. Each fit draws 5000 posterior samples after 5000 burn-in iterations. We report the root mean squared error, bias, 95% credible-interval coverage, and posterior variance of the estimated CATE.

We investigate the performance of the Bayesian fusion forest in three separate simulation setups.

1) **Unmeasured confounding and between-source heterogeneity.** We investigate how the bias and RMSE of the fusion forest behave as the unmeasured confounding and the between-source heterogeneity grow. We vary the confounding strength $\lambda_u \in \{0, 0.5, 1, 1.5, 2\}$ at fixed $\lambda_d = 1$, and the heterogeneity $\lambda_d \in \{0, 0.5, 1, 1.5, 2\}$ at fixed $\lambda_u = 1$.

2) **Comparison with a black-box competitor.** We investigate how the fusion forest compares with a flexible deep-learning method. We compare it with a deep neural network for the accelerated failure time model (Norman et al., 2024), tuned by 10-fold cross-validation. We estimate a causal effect using an S- and T-learner (Künzel et al., 2019), each fitted on the trial alone and on a naive pool of both sources. We obtain confidence intervals from 100 bootstrap resamples. We fix $\lambda_d = \lambda_u = 1$. The deep-learning competitor cannot handle interval censoring, so for this experiment we right-censor the real-world data at circa 35%.

3) **Increasing covariate dimension.** We investigate how the precision gained by borrowing from the real-world data trades off against the cost of searching a larger covariate space, and whether fusion remains worthwhile in higher covariate space dimensions. We hold the sample sizes fixed and grow the number of covariates from $p = 5$ to $p = 500$ at $\lambda_d = \lambda_u = 1$. Only five covariates carry signal, so the remaining covariates are pure noise and the problem grows sparser. We draw the covariates in independent blocks of ten with the AR(1) correlation $\rho = 0.3$ within each block. We also fit an oracle variant given only the five active covariates. It marks the best performance attainable when the active covariates are known.

We additionally test the gain from modelling the error distribution nonparametrically. We compare the hierarchical Dirichlet process mixture with a single Gaussian and a pooled Dirichlet

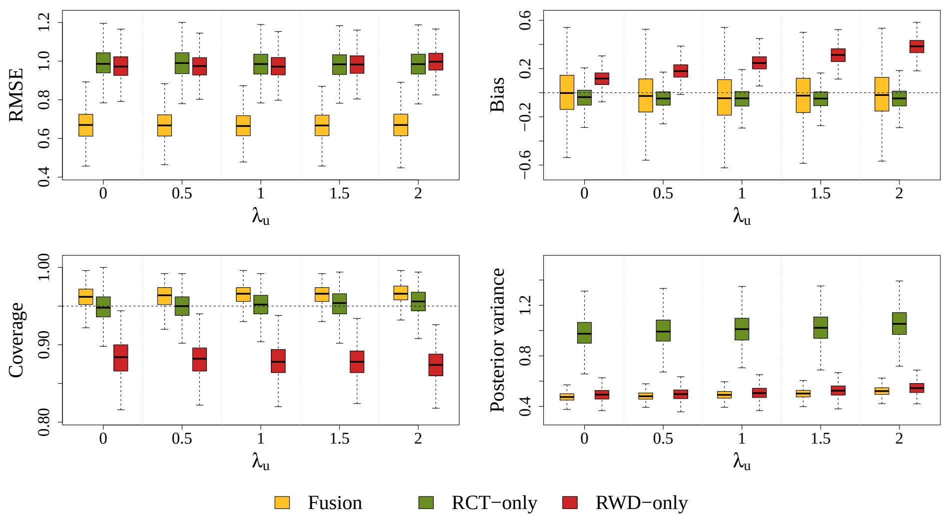


Figure 1: CATE metrics under varying unmeasured confounding strength $\lambda_u$, with the RWD prognostic deviation held at $\lambda_d = 1$. From top-left to bottom-right: pointwise RMSE, mean bias, 95% credible-interval coverage, and posterior variance of $\hat{\tau}(x)$, averaged over the evaluation set and across simulation replicates. Dashed reference lines indicate zero bias and nominal coverage.

process mixture, both of which force the two sources to share one error law. The nonparametric model is robust across error laws and improves on both alternatives (Supplementary Materials S.4).

### 3.2 Simulation results

We first vary the confounding strength $\lambda_u$ and hold the between-source heterogeneity fixed at $\lambda_d = 1$. Figure 1 reports the results. The real-world-only estimate is unbiased at $\lambda_u = 0$. Its bias then grows roughly linearly with $\lambda_u$, to circa 0.4 at $\lambda_u = 2$. The trial-only estimate is unaffected by $\lambda_u$. The Bayesian fusion forest stays essentially unbiased across the whole range. The Bayesian fusion forest also attains a lower RMSE than the trial-only baseline at every level. Coverage is near nominal for the fusion and trial-only estimates. The real-world-only estimate loses coverage as its bias grows. The fusion posterior variance is roughly half that of the trial-only baseline throughout. This reflects the precision gained by borrowing from the larger real-world sample.

We report the results for varying between-source heterogeneity $\lambda_d$ at fixed confounding $\lambda_u = 1$ in Supplementary Materials S.4. All three estimators are essentially flat in $\lambda_d$. The fusion and trial-only estimates stay unbiased and at nominal coverage. The real-world-only estimate keeps the bias from the fixed confounding. RMSE and posterior variance are stable across $\lambda_d$.

We compare the deep neural network competitors at $\lambda_d = \lambda_u = 1$ in Figure 2. The Bayesian fusion forest attains the lowest RMSE of all methods. The S-learners show undercoverage, a consequence of their narrow bootstrap intervals. The pooled T-learner shows overcoverage, together with a large variance. The fusion forest attains near-nominal coverage. We give the full results across the $(\lambda_d, \lambda_u)$ grid in Supplementary Materials S.4.

We finally examine robustness to the covariate dimension. Figure 3 reports the CATE metrics against $p$. The fusion forest stays essentially unbiased across the whole range, as does the trial-only baseline, while the real-world-only estimate retains its confounding bias. All estimators lose precision as $p$ grows, but at very different rates. The trial-only posterior variance increases several-fold, whereas the fusion variance grows only mildly, and the fusion attains the lowest RMSE of the feasible estimators at every $p$. The fusion holds near-nominal coverage throughout, while the trial-only intervals turn conservative as their variance increases. The efficiency of the fusion relative to the trial-only baseline confirms that the gain persists. The RMSE ratio rises from 0.68 at $p = 10$ to 0.81 at $p = 500$, yet stays below one throughout. The posterior-variance ratio stays near one-half (Supplementary Materials S.4). Because both estimators are unbiased, this advantage is essentially a variance effect. The fusion borrows the larger real-world sample to control a variance the trial alone cannot.

These results show that the fusion forest is robust to both nuisances. The fusion forest corrects for the unmeasured confounding in the real-world data and stays unbiased as the confounding strengthens. It accommodates the prognostic heterogeneity between the sources and is unaffected as the heterogeneity grows. It also exploits the larger real-world sample and achieves lower

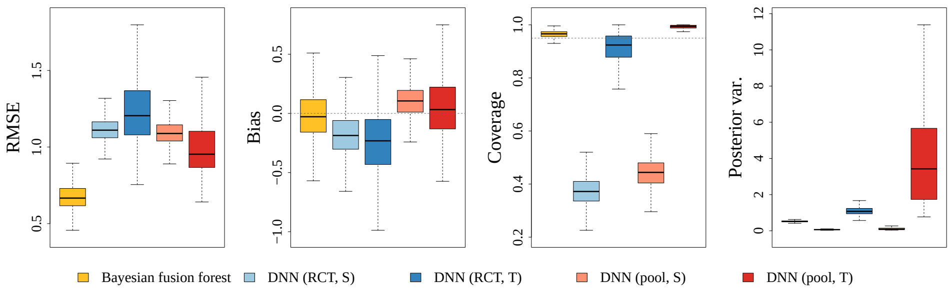


Figure 2: CATE metrics for the Bayesian fusion forest and four deep neural networks (DNN) competitors at $\lambda_d = \lambda_u = 1$. Each applies the DNN as an S- or T-learner, on the RCT alone or a naive pool of both sources. From top-left to bottom-right: RMSE, bias, coverage, and variance of $\hat{\tau}(x)$. Dashed reference lines indicate zero bias and nominal coverage.

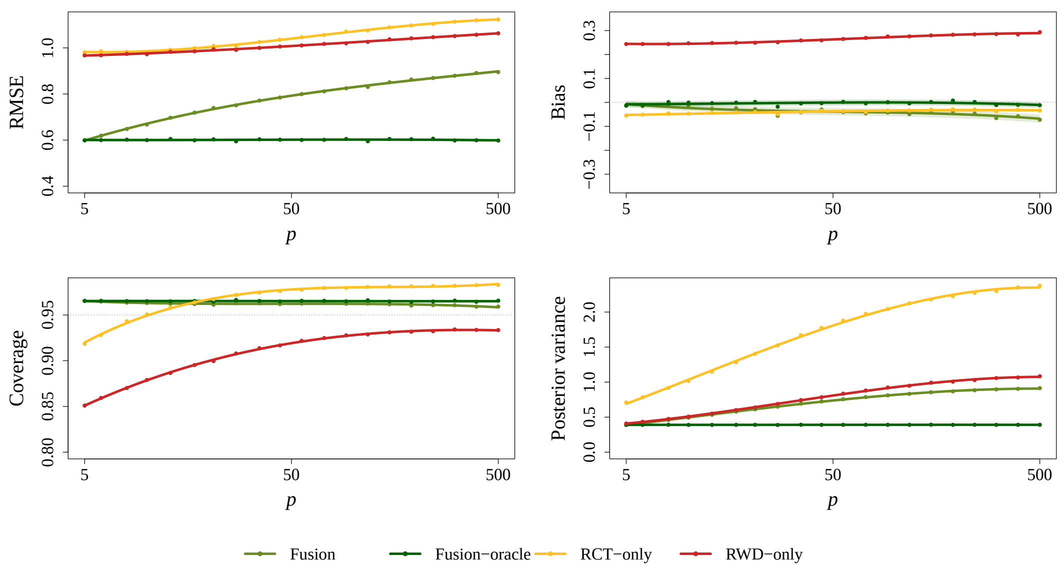


Figure 3: CATE metrics as the number of covariates $p$ grows from 5 to 500, with $n_1 = 150$, $n_0 = 350$, and $\lambda_d = \lambda_u = 1$; five covariates are active and the remaining $p - 5$ are noise. From top-left to bottom-right: RMSE, bias, 95% credible-interval coverage, and posterior variance of $\hat{\tau}(x)$. The fusion oracle is the fusion forest given only the five active covariates, and traces the achievable floor. Dashed reference lines mark zero bias and nominal coverage.

RMSE and posterior variance than the trial-only baseline at every grid point.

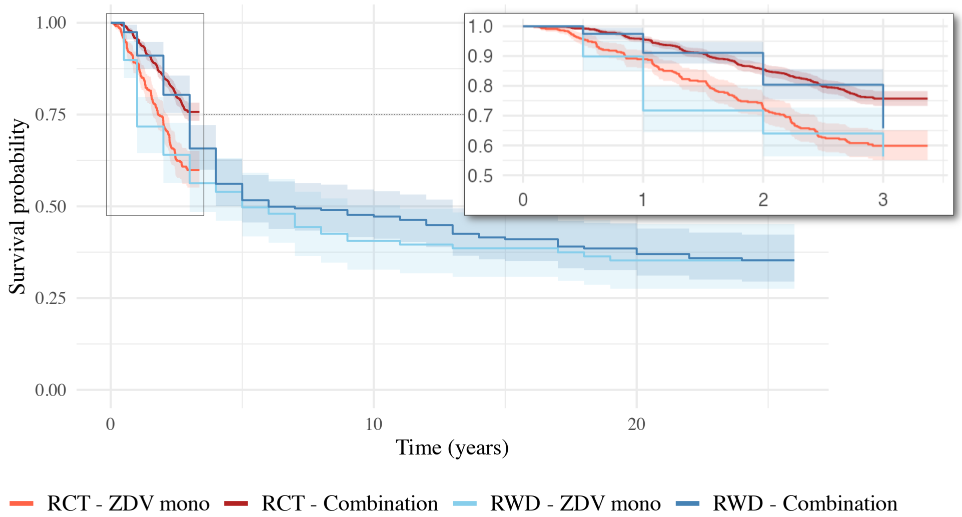


Figure 4: Kaplan–Meier curves of event-free survival, by data source (RCT, RWD) and treatment group (ZDV monotherapy and combination). The main panel spans the full cohort follow-up of circa 25 years; the inset zooms to the shorter trial window. The real-world data curves are approximate due to interval censoring.

# 4 Application: survival after HIV

We study the effect of a combination antiretroviral therapy versus zidovudine (ZDV) monotherapy on event-free survival in HIV. We combine the AIDS Clinical Trials Group Protocol 175 (ACTG 175) (Hammer et al., 1996) with the Multicenter AIDS Cohort Study (MACS) (Kaslow et al., 1987). Treatment in ACTG 175 is randomised. We pool the three combination regimens in ACTG 175 into one comparator and contrast it with ZDV monotherapy. We restrict both sources to a common population: men with baseline CD4 count of 200 to 500 cells/mm$^3$. This alignment makes the fusion credible, since the two sources then approximate one population. We adjust for six baseline covariates: age, CD4, CD8, calendar year, race, and years of prior antiretroviral therapy. We summarise the aligned cohort in Supplementary Materials S.5. The aligned cohort contains 1771 trial patients and 373 real-world data patients. The trial follows patients for a maximum of circa three years: too short to reach median event-free survival. The MACS study follows them for circa 25 years and reaches a median event-free survival of five to six years. This longer follow-up adds robustness for the late treatment effect where the trial is uninformative. MACS records outcomes at annual visits. The data give event times only to the calendar year, so the events are interval-censored at annual resolution. Figure 4 shows the Kaplan–Meier survival curves.

We fit a Bayesian fusion forest to the combined data with default parameter settings. We compare it with an AFT Bayesian causal forest fitted to the trial alone. We compute the average effect with a source-specific Bayesian bootstrap. The source weights follow a Dirichlet prior proportional to the sample sizes. The fusion estimate of the average acceleration factor is 1.65 (95% CrI: 1.43 to 1.90). The trial-only baseline is close at 1.70 (95% CrI: 1.51 to 1.92). Combination therapy stretches the event-free survival time scale by a factor of 1.65 relative to monotherapy. Equivalently, the median event-free survival time under combination therapy is 1.65 times the median under monotherapy.

The Bayesian fusion forest estimates each patient's effect far more precisely than the trial alone. Figure 5 shows the subject-level acceleration factor for every patient, ordered by posterior mean.

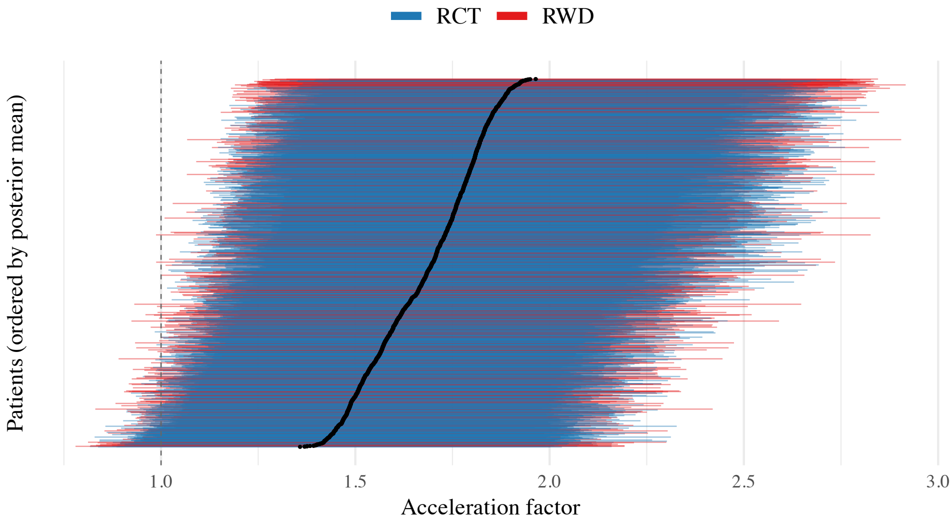


Figure 5: Subject-level acceleration factor for every patient in the trial-aligned cohort, from the Bayesian fusion forest, ordered by posterior mean. Vertical bars are pointwise 95% credible intervals; colour indicates the data source (RCT, RWD); the dashed line at 1 marks no effect.

The estimates lie above one, so combination helps almost everyone. The size of the gain differs across patients. The fusion intervals are far narrower than the trial-only intervals. The average 95% credible-interval width falls from 3.16 to 1.21 on the acceleration-factor scale, a reduction of 62% (Supplementary Materials S.5). The fusion sharpens the individual estimates. This contrasts with the average effects presented above: the fusion interval there is slightly wider, because the Bayesian bootstrap propagates uncertainty about the target population's covariate distribution.

We compare the fusion estimates with a flexible deep-learning competitor on the cohort. We fit a deep accelerated failure time network as an S- and a T-learner, on the trial alone and on a naive pool of both sources (Supplementary Materials S.5). We report in Table 1 the only two quantities a point predictor yields on the causal scale: the mean width of the per-patient interval and the share of patients at least 95% certain to benefit. Supplementary Materials S.5 plot the per-patient acceleration factor and its uncertainty for each method. The deep neural network reaches a different conclusion for each combination of learner and pooling. The share certain to benefit ranges from 7.8% under the pooled T-learner to 96.6% under the pooled S-learner, and the interval width ranges just as widely. No principled rule selects among these configurations. The pooled S-learner nearly matches the fusion (96.6% against 96.4%), but the equally defensible pooled T-learner reverses it. Its probability of benefit is in any case a bootstrap proportion, not a posterior probability. The S-learners give the narrowest intervals of all. The simulation showed such intervals undercover, so their narrowness reflects overconfidence rather than precision. The Bayesian fusion forest instead gives a single coherent answer, with calibrated intervals and a posterior probability of benefit for every patient.

We investigate which patient characteristics drive the variation in benefit. We summarise the treatment-effect forest $\tau$ by the posterior projection, with a single regression tree as the second-stage model (CART; Breiman et al., 1984). Figure 6 shows a depth-three tree on the six covariates. It splits on CD8 count and race. The other covariates do not enter. The acceleration factor rises with CD8, from circa 1.5 in the lowest group to circa 1.8 in the highest. White patients gain slightly more than non-white within each CD8 band. Every leaf exceeds 1.4, so all subgroups benefit. The treatment benefit is largest at high CD8.

| Method | Mean 95% width (AF scale) | $\geq$ 95% certain of benefit (%) |
|---|---|---|
| Bayesian fusion forest | 1.21 | 96.4 |
| Trial-only causal forest | 3.16 | 36.8 |
| Deep network, trial-only, S | 0.58 | 77.5 |
| Deep network, trial-only, T | 1.79 | 25.8 |
| Deep network, pooled, S | 0.88 | 96.6 |
| Deep network, pooled, T | 2.34 | 7.8 |

Table 1: Comparison with a deep-learning competitor over the combined RCT + MACS cohort ($n = 2144$). "Mean 95% width" is the average width of the per-patient interval on the acceleration-factor scale. The last column is the percentage of patients at least 95% certain to benefit, that is with $\mathbb{P}(\tau(x) > 0 \mid \text{data}) \geq 0.95$; for the deep network this probability is a bootstrap proportion rather than a posterior. The deep network is fitted as an S- or T-learner on the trial alone or on a naive pool of both sources; the setup and the subject-level estimates are in Supplementary Materials S.5.

We inspect the posterior probability of treatment benefit $\mathbb{P}(\tau(x_i) > 0 \mid \text{data})$ for each patient (Henderson et al., 2020). We bin the patients by posterior probability of benefit and report the fraction in each bin. We compute it over the trial population, under the fusion and trial-only fits, so the columns are comparable (Table 2). We find strong evidence of benefit for almost every patient. The trial alone is far less certain, with only 38% above 0.95 and a quarter of patients inconclusive. The Bayesian fusion forest gains this certainty at the conditional level even though its average effect is slightly smaller.

We investigate the between-source heterogeneity in baseline prognosis. We summarise the deviation forest $d(x)$ by a projection on a linear model. The deviation forest measures how the baseline prognosis differs between the trial and the real-world data. We project $d(\cdot)$ onto the covariates by least squares and present the results in Table 3. The gap is dominated by a constant shift. The intercept is 1.70 (95% CrI: 1.31 to 2.10), so the observational baseline sits well above the trial. Among the covariates, only prior antiretroviral years moves the gap, by 0.27 (95% CrI: $-0.01$ to 0.55). Age, CD4, CD8, calendar year, and race have coefficients near zero, with intervals covering zero. The baseline prognosis therefore differs mainly through a large average shift. This shift likely reflects population and measurement differences which the six covariates do not capture.

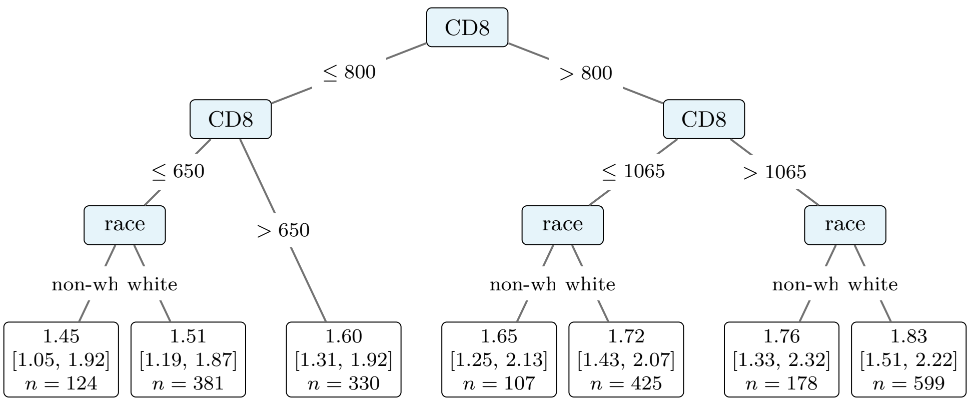


Figure 6: Regression-tree summary of the acceleration factor $\exp(\tau(x))$ from the fusion forest. We fit the tree to the posterior-mean log-time CATE on the six baseline covariates. Each leaf gives the within-leaf average acceleration factor, as a posterior mean and 95% credible interval.

| $\mathbb{P}(\tau(x_i) > 0 \mid \text{data})$ | Fusion | Trial-only |
|---|---|---|
| $(0.99, 1]$ | 0.767 | 0.180 |
| $(0.95, 0.99]$ | 0.206 | 0.198 |
| $(0.75, 0.95]$ | 0.028 | 0.364 |
| $(0.25, 0.75]$ | 0.000 | 0.245 |
| $[0, 0.25]$ | 0.000 | 0.014 |

Table 2: Posterior probability of benefit over the trial population. Higher bins give stronger evidence of benefit, the lowest evidence of harm. Entries are fractions of patients.

| Term | Coefficient | 95% CrI |
|---|---|---|
| Intercept | 1.701 | $(1.312, 2.099)$ |
| Age | $-0.003$ | $(-0.020, 0.015)$ |
| CD4 | 0.001 | $(-0.001, 0.002)$ |
| CD8 | 0.000 | $(-0.000, 0.000)$ |
| Calendar year | $-0.032$ | $(-0.256, 0.194)$ |
| Race | 0.071 | $(-0.304, 0.443)$ |
| Prior ART, y | 0.269 | $(-0.008, 0.550)$ |

Table 3: Linear projection of the deviation forest $d(x)$ onto the covariates: posterior mean coefficient and 95% credible interval on the log-transformed time scale.

Combination therapy prolongs event-free survival by a factor of circa 1.65. We identify benefit for nearly every patient, far more than the trial alone. This near-universal advantage echoes earlier ACTG 175 analyses. The original trial (Hammer et al., 1996) and later re-analyses on survival-type endpoints (Lee and Kim, 2026) all favour combination therapy over monotherapy. The Buckley–James Q-learner (Lee and Kim, 2026) likewise recommends it for over 96% of participants. We found that the treatment effect varies with CD8 count and race. This agrees with evidence that the benefit of combination therapy varies with baseline risk (Kennedy et al., 2023). The baseline prognosis differs between the sources, mainly through a large average shift and in part through its dependence on prior antiretroviral exposure. The fusion delivers this certainty with calibrated uncertainty that a black-box predictor does not provide.

# 5 Discussion

We combined a randomised trial with real-world data to estimate heterogeneous treatment effects on survival outcomes, without assuming the real-world source is unconfounded. The simulation study and the HIV application demonstrate its advantages over a single-source analysis. The Bayesian fusion forest remains unbiased under varying levels of unmeasured confounding. It reduces the posterior variance of a trial-only analysis. In the application, we recovered a benefit of combination therapy that the survival curves of the real-world data alone did not reveal. We established benefit for nearly every patient where the trial alone was inconclusive.

Our framework implicitly assumes the acceleration factor $\exp(\tau(X))$ is time-invariant: treatment rescales the survival time by a factor that depends on the covariates but not on time. This assumption may be inappropriate when treatment effects unfold over time, such as delayed-onset benefits or waning effects. Flexible parametric AFT models relax this assumption by letting the acceleration factor vary with time (Crowther et al., 2023). The causal interpretation of time-dependent acceleration factors has been formalised (Brathovde et al., 2026). A natural extension replaces $\tau(X)$ with a generalised Bayesian tree ensemble $\tau(X, t)$.

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## Conflict of interest

None declared.

## Data availability

The software implementation is available on GitHub at https://github.com/tijn-jacobs/FusionForests. *CRAN release will follow soon.* The ACTG 175 trial data are available in the `speff2trial` R package (Juraska et al., 2022). The MACS data can be requested through the MWCCS cohort website.

## References


Brathovde, M., Putter, H., Valberg, M., and Post, R. A. J. (2026). The causal interpretation of acceleration factors. arXiv:2409.01983.

Breiman, L., Friedman, J. H., Olshen, R. A., and Stone, C. J. (1984). *Classification and Regression Trees*. Chapman and Hall/CRC., 1 edition.

Chen, M.-H. and Ibrahim, J. G. (2000). Power prior distributions for regression models. *Statistical Science*, 15(1):46–60.

Chipman, H. A., George, E. I., and McCulloch, R. E. (1998). Bayesian CART model search. *Journal of the American Statistical Association*, 93(443):935–948.

Chipman, H. A., George, E. I., and McCulloch, R. E. (2010). BART: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1):266–298.

Crowther, M. J., Royston, P., and Clements, M. (2023). A flexible parametric accelerated failure time model and the extension to time-dependent acceleration factors. *Biostatistics*, 24(3):811–831.

Dahabreh, I. J., Robertson, S. E., Tchetgen, E. J., Stuart, E. A., and Hernán, M. A. (2019). Generalizing causal inferences from individuals in randomized trials to all trial-eligible individuals. *Biometrics*, 75(2):685–694.

Dimitriou, E., Fong, E., Tarp, J. M., Diaz-Ordaz, K., and Lehmann, B. (2026). Causal-ICM: A data fusion framework for heterogeneous treatment effect estimation with multi-task Gaussian processes. In *Proceedings of Machine Learning Research*, volume 323, pages 1–29.

Dorie, V., Hill, J., Shalit, U., Scott, M., and Cervone, D. (2019). Automated versus do-it-yourself methods for causal inference: Lessons learned from a data analysis competition. *Statistical Science*, 34(1):43–68.

Foster, J. C., Taylor, J. M. G., and Ruberg, S. J. (2011). Subgroup identification from randomized clinical trial data. *Statistics in Medicine*, 30(24):2867–2880.

Hahn, P. R., Murray, J. S., and Carvalho, C. M. (2020). Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects. *Bayesian Analysis*, 15(3):965–1056.

Hammer, S. M., Katzenstein, D. A., Hughes, M. D., Gundacke, H., Schooley, R. T., Haubrich, R., Henry, W., Lederman, M., Phair, J., Niu, M., Hirsch, M., and Merigan, T. (1996). A trial comparing nucleoside monotherapy with combination therapy in HIV-infected adults with CD4 cell counts from 200 to 500 per cubic millimeter. *New England Journal of Medicine*, 335(15):1081–1090.

Henderson, N. C., Louis, T. A., Rosner, G. L., and Varadhan, R. (2020). Individualized treatment effects with censored data via fully nonparametric Bayesian accelerated failure time models. *Biostatistics*, 21(1):50–68.

Hernán, M. A. (2010). The hazards of hazard ratios. *Epidemiology*, 21(1):13–15.

Hill, J. L. (2011). Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240.

Hobbs, B. P., Sargent, D. J., and Carlin, B. P. (2012). Commensurate priors for incorporating historical information in clinical trials using general and generalized linear models. *Bayesian Analysis*, 7(3):639–674.

Jacobs, T. (2026). ShrinkageTrees: An R Package for Bayesian Tree Ensembles for Survival Analysis and Causal Inference.

Jacobs, T., van Wieringen, W. N., and van der Pas, S. L. (2025). Horseshoe forests for high-dimensional causal survival analysis. Accepted for publication in Bayesian Analysis.

Juraska, M., Gilbert, P. B., Lu, X., Zhang, M., Davidian, M., and Tsiatis, A. A. (2022). *speff2trial: Semiparametric Efficient Estimation for a Two-Sample Treatment Effect*. R package version 1.0.5.

Kabata, D., Henderson, N. C., and Varadhan, R. (2026). Quantifying uncertainty of individualized treatment effects in right-censored survival data: A comparison of Bayesian additive regression trees and causal survival forest. *Health Services and Outcomes Research Methodology*, 26:87–107.

Kallus, N., Puli, A. M., and Shalit, U. (2018). Removing hidden confounding by experimental grounding. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, NIPS'18, page 10911–10920, Red Hook, NY, USA. Curran Associates Inc.

Kaslow, R. A., Ostrow, D. G., Detels, R., Phair, J. P., Polk, B. F., Rinaldo, C. R., and STUDY, F. T. M. A. C. (1987). The Multicenter AIDS Cohort Study: rationale, organization, and selected characteristics of the participants. *American Journal of Epidemiology*, 126(2):310–318.

Kennedy, E. H., Balakrishnan, S., and Wasserman, L. A. (2023). Semiparametric counterfactual density estimation. *Biometrika*, 110(4):875–896.

Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165.

Lee, J. and Kim, J.-M. (2026). Counterfactual survival Q-learning via Buckley–James boosting, with applications to ACTG 175 and CALGB 8923. *Journal of the Royal Statistical Society Series C: Applied Statistics*. Advance access.

Mao, G., Yang, S., and Wang, X. (2025). Statistical inference for heterogeneous treatment effect with right-censored data from synthesizing randomized clinical trials and real-world data. *Biometrics*, 81(4):ujaf131.

Neuenschwander, B., Capkun-Niggli, G., Branson, M., and Spiegelhalter, D. J. (2010). Summarizing historical information on controls in clinical trials. *Clinical Trials*, 7(1):5–18.

Norman, P. A., Li, W., Jiang, W., and Chen, B. E. (2024). deepAFT: A nonlinear accelerated failure time model with artificial neural network. *Statistics in Medicine*, 43(19):3689–3701.

Robins, J. M., Rotnitzky, A., and Scharfstein, D. O. (2000). Sensitivity analysis for selection bias and unmeasured confounding in missing data and causal inference models. In Halloran, M. E. and Berry, D., editors, *Statistical Models in Epidemiology, the Environment, and Clinical Trials*, pages 1–94, New York, NY. Springer New York.

Ročková, V. and van der Pas, S. (2020). Posterior concentration for Bayesian regression trees and forests. *The Annals of Statistics*, 48(4):2108–2131.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701.

Rubin, D. B. (1981). The Bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134.

Shyr, C., Ren, B., Patil, P., and Parmigiani, G. (2025). Multi-study $R$-learner for estimating heterogeneous treatment effects across studies using statistical machine learning. *Biostatistics*, 26(1):kxaf040.

Stuart, E. A., Cole, S. R., Bradshaw, C. P., and Leaf, P. J. (2011). The use of propensity scores to assess the generalizability of results from randomized trials. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 174(2):369–386.

Sun, J. (2006). *The Statistical Analysis of Interval-Censored Failure Time Data*. Statistics for Biology and Health. Springer, New York.

Swindell, W. R. (2009). Accelerated failure time models provide a useful statistical framework for aging research. *Experimental Gerontology*, 44(3):190–200.

Tanner, M. A. and Wong, W. H. (1987). The calculation of posterior distributions by data augmentation. *Journal of the American Statistical Association*, 82(398):528–540.

Teh, Y. W., Jordan, M. I., Beal, M. J., and Blei, D. M. (2006). Hierarchical Dirichlet processes. *Journal of the American Statistical Association*, 101(476):1566–1581.

Thal, D. R. C. and Finucane, M. M. (2023). Causal methods madness: Lessons learned from the 2022 ACIC competition to estimate health policy impacts. *Observational Studies*, 9(3):3–27.

VanderWeele, T. J. and Ding, P. (2017). Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274.

Woody, S., Carvalho, C. M., and Murray, J. S. (2021). Model interpretation through lower-dimensional posterior summarization. *Journal of Computational and Graphical Statistics*, 30(1):144–161.

Yang, M., Dunson, D. B., and Baird, D. (2010). Semiparametric Bayes hierarchical models with mean and variance constraints. *Computational Statistics and Data Analysis*, 54(9):2172–2186.

Yang, S., Gao, C., Zeng, D., and Wang, X. (2023). Elastic integrative analysis of randomised trial and real-world data for treatment heterogeneity estimation. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 85(3):575–596.

Yang, S., Liu, S., Zeng, D., and Wang, X. (2025). Data fusion methods for the heterogeneity of treatment effect and confounding function. *Bernoulli*, 31(4):2987–3012.

Ye, X., Yang, S., Wang, X., and Liu, Y. (2025). Integrative analysis of high-dimensional RCT and RWD subject to censoring and hidden confounding. *Lifetime Data Analysis*, 31:473–497.

Zhou, T. and Ji, Y. (2021). Incorporating external data into the analysis of clinical trials via Bayesian additive regression trees. *Statistics in Medicine*, 40(28):6421–6442.